

Max Larimer

Final Assignment

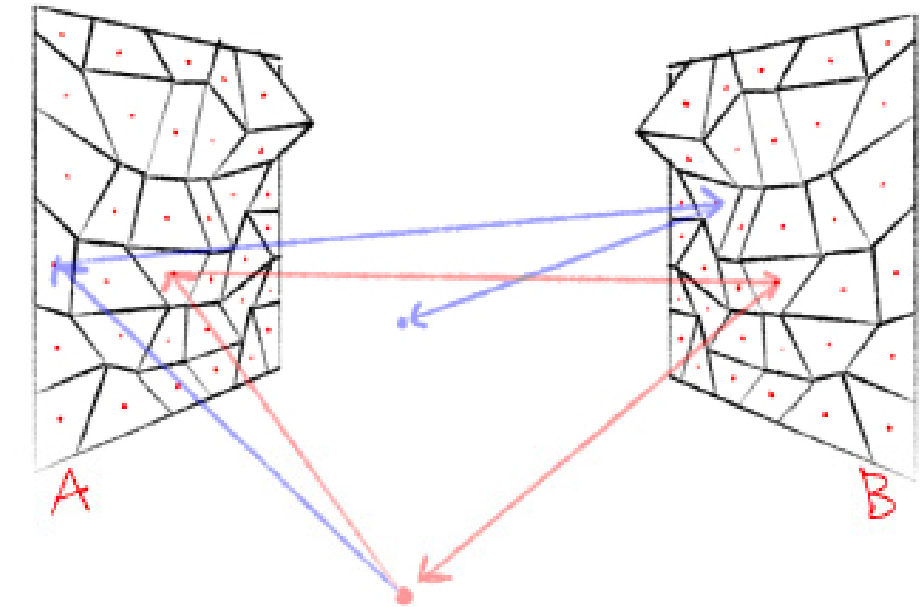
ANNOTATED SKETCH

Grasshopper Sketch

In the first step of this project, we created an annotated sketch that starts to think about parametric modeling. In this sketch, I have started to think about how I would like to find the reflection points on separate surfaces.

In the bottom left, I have created a hierarchy for the different panels so that I could find the normals of each point.

In the 3d sketch, I was visualizing how setting different points would create the facets by orienting the normals to reflect to the other point.



W

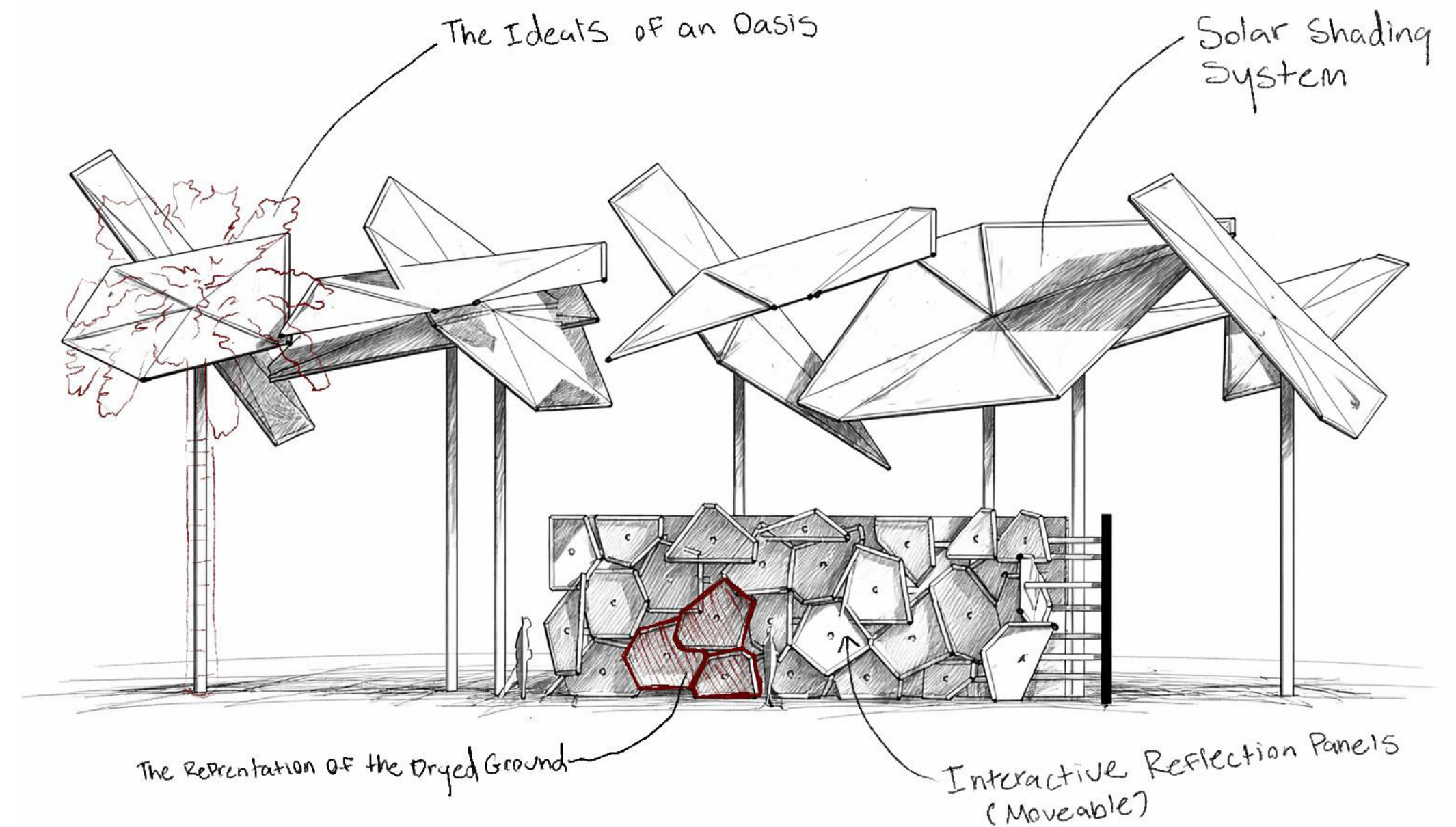
w	w	w	w	w	w
0.0	0.1	0.2	0.3	0.4	0.5
1.0	1.2	1.3	1.4	1.5	1.6
2.0	2.1	2.2	2.3	2.4	2.5
3.0	3.1	3.2	3.3	3.4	3.5
4.0	4.1	4.2	4.3	4.4	4.5

H

Vector = Point A - Point B
→ Vector = Point A → Plane A2.1 → Plane B2.1 → Point B
* How do you Randomize?

THE CONCEPT

While this sketch came much later, the concept of the pavilion remains important. The concept behind this pavilion was to create a shaded canopy "forest" in the desert and then once arriving at the forest, there would be interactive reflection panels that allowed the users to direct where these reflections are directed towards. The idea was to create an element of oasis while representing reflection and perspective.



DESIGNING IN GRASSHOPPER

Scripting

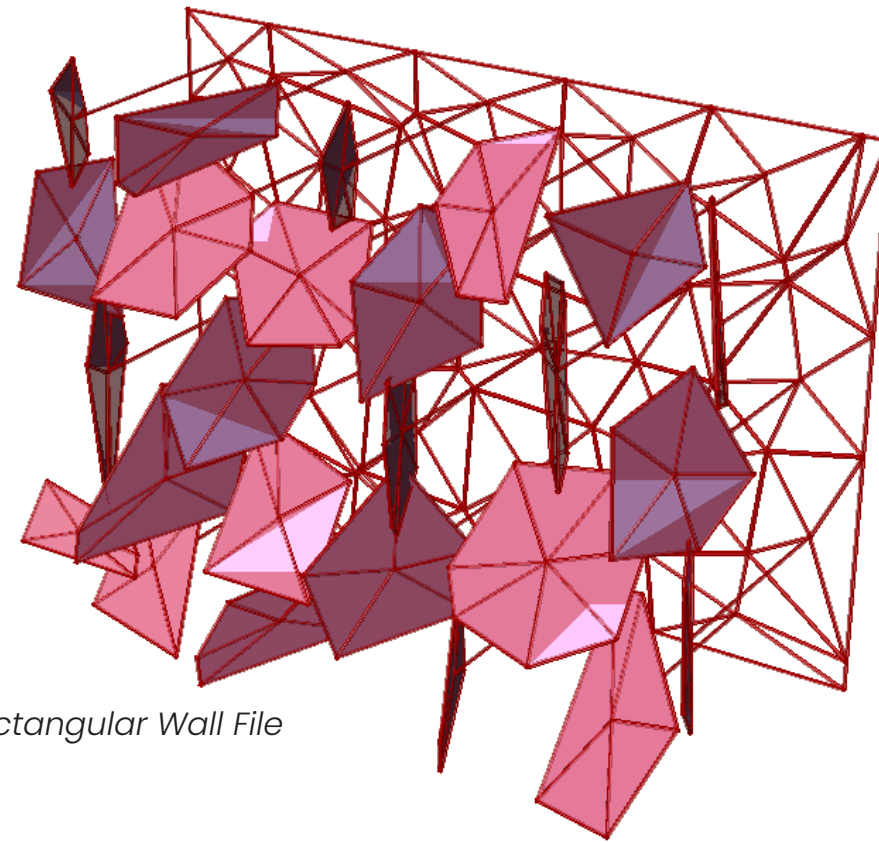
Moving from the sketch to modeling in grasshopper, I began working through iterations of different panelized surfaces, because my site was in the desert, The vernoi pattern became the concept of how I divided the surface up into different shapes. I began developing a systems that could be used to create inhabit-able spaces. This led to 2 different scripts:

The first script that was made was only able to be assigned to rectangular planes but in this case was able to act as the walls or barriers of the pavilion.

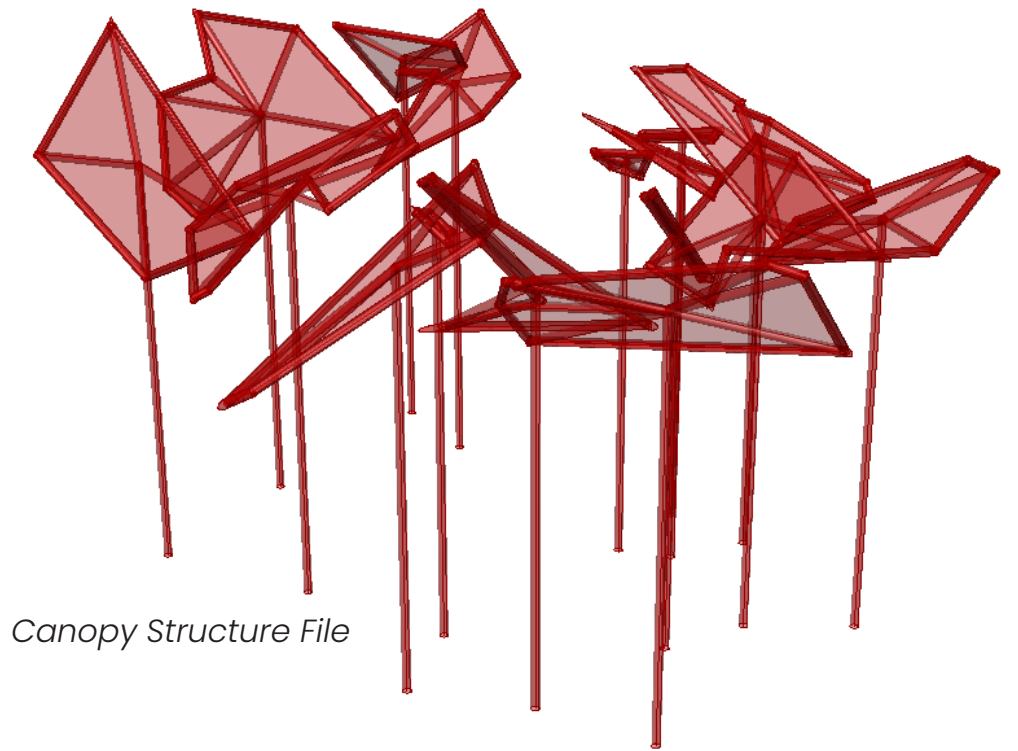
The second script was developed to act as a roof canopy system, the main element of this script was the ability to apply the vernoi pattern to a non rectangular planes.

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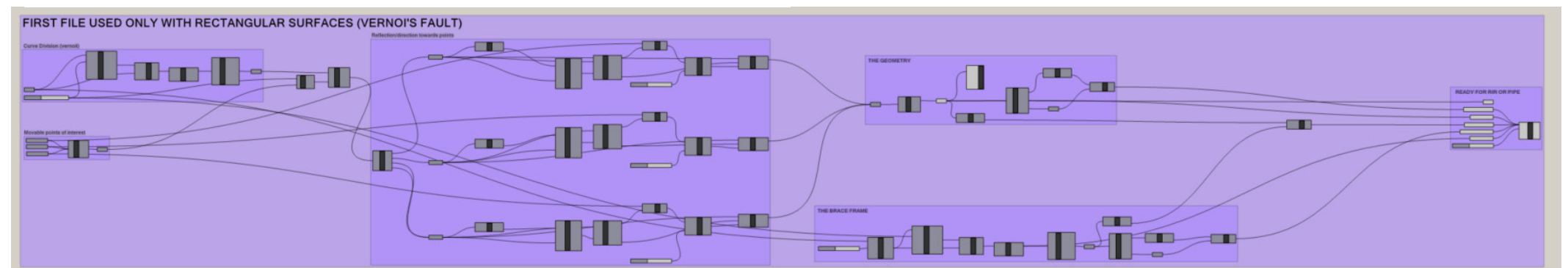
ARCH 565



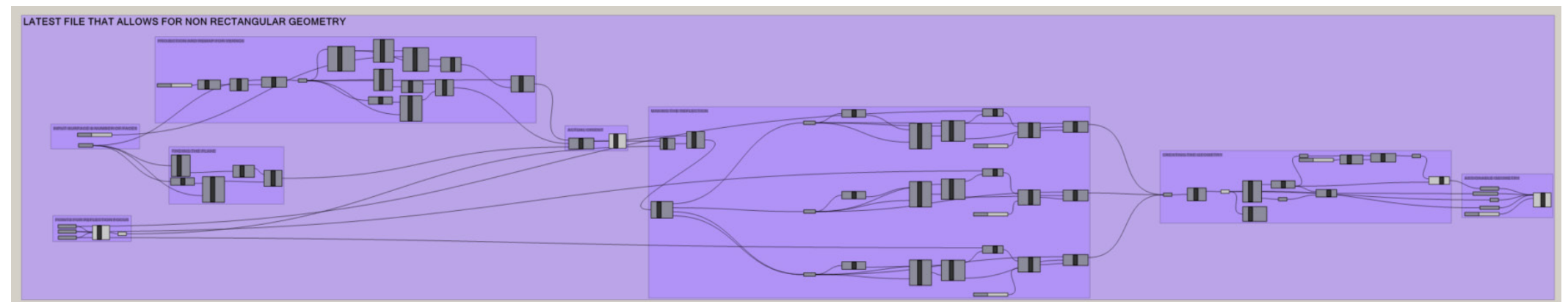
Rectangular Wall File



Canopy Structure File



Rectangular Wall File



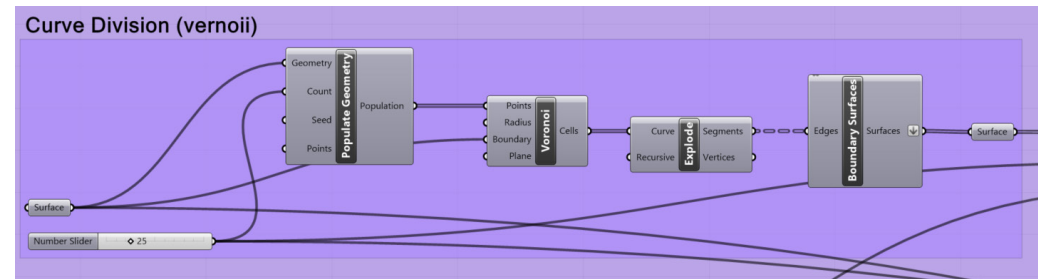
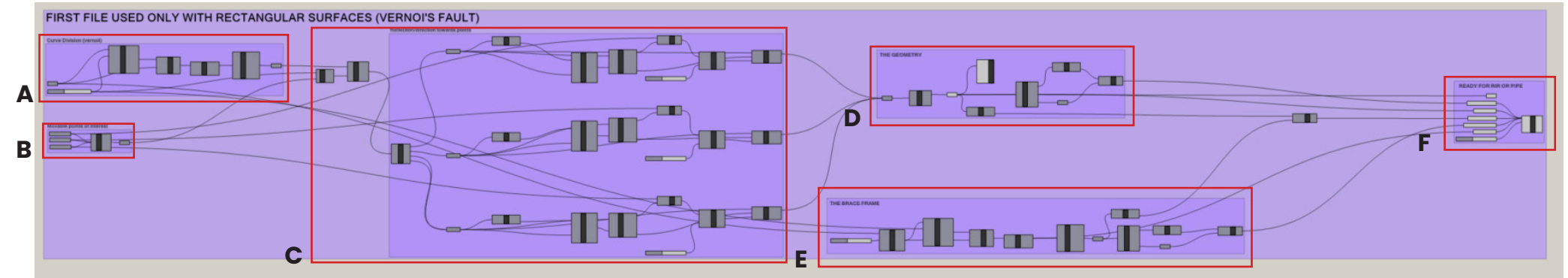
Canopy Structure File

WALL SCRIPT

Rectangular Planes

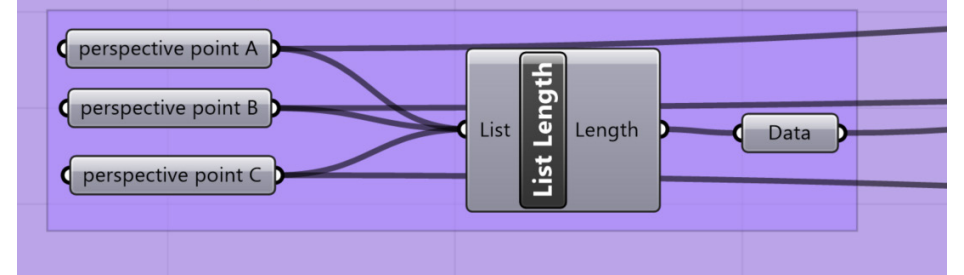
FALL 2025

ARCH 565

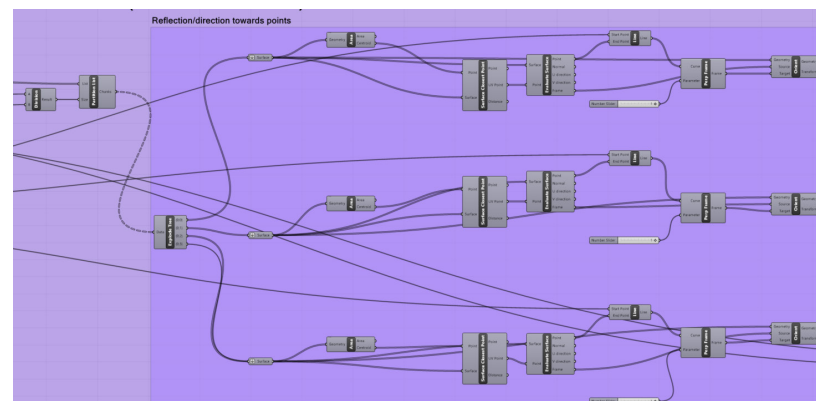


A The beginning of the script starts with the input surface and then works through the steps of populating the surface with the vernoi pattern.

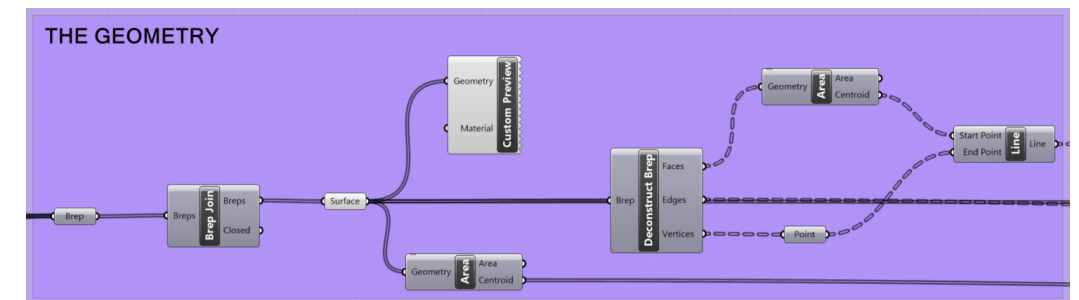
Movable points of interest



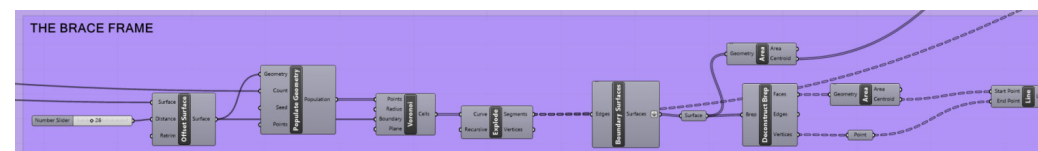
B This input correlates to the points that dictate the direction of the surface normals. These points can be placed by assignable points to direct to certain places.



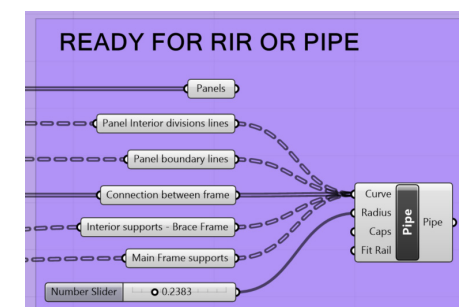
C This sequence starts by randomly assigning panels to the respective points and then orients the surfaces to reflect to the corresponding point.



D Once the surfaces are oriented, this process begins pulling the elements of the geometry out of the surfaces to create the structural systems.



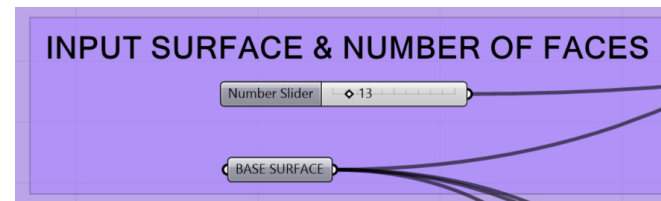
E This portion of the script creates the brace frame that can be embedded in the wall or surface to provide support to the surfaces that are protruding.



F The output consists of lines for the boundary of the surfaces, interior division to the center of each panel, connecting lines between frame and panel, and the brace.

Horizontal non - Rectangular

ARCH 565



Movable points of interest

The diagram illustrates a system architecture for 'Movable points of interest'. On the left, three yellow rounded rectangles represent 'perspective point A', 'perspective point B', and 'perspective point C'. Arrows from these points converge into a central grey rounded rectangle labeled 'List Length'. This central rectangle is divided into two sections: 'List' on the left and 'Length' on the right. An arrow points from the 'Length' section to a grey rounded rectangle labeled 'Data'. Finally, an arrow points from the 'Data' rectangle to the right edge of the diagram.

FINDING THE PLANE

```

graph LR
    A[Decomposed Edge  
Edges  
Vertices] --> D[Plane Client Plane  
Point  
Point  
Surface  
Dist  
C/P Point  
Distance]
    B[Geometry  
Area  
Perim] --> D
    D --> C[Plane H  
Plane  
Vertices]
    C --> E[Plane Origin  
Origin  
Plane]
  
```

[illegible]

The diagram, titled "MINING THE REPLICATION", illustrates a network topology for data replication. It features a central horizontal backbone of nodes. From this backbone, several branches extend outwards, each ending in a cluster of nodes. These clusters are interconnected by a series of curved lines, representing the replication paths. The diagram is set against a light blue background with a grid pattern.

[illegible]

ASSIGNABLE GEOMETRY

STRUCTURE DOWN

Panel interior division lines

Panels

Panel boundary lines

Number Slider: 0.0000

Curve
Radius
Capis
Fit Rat

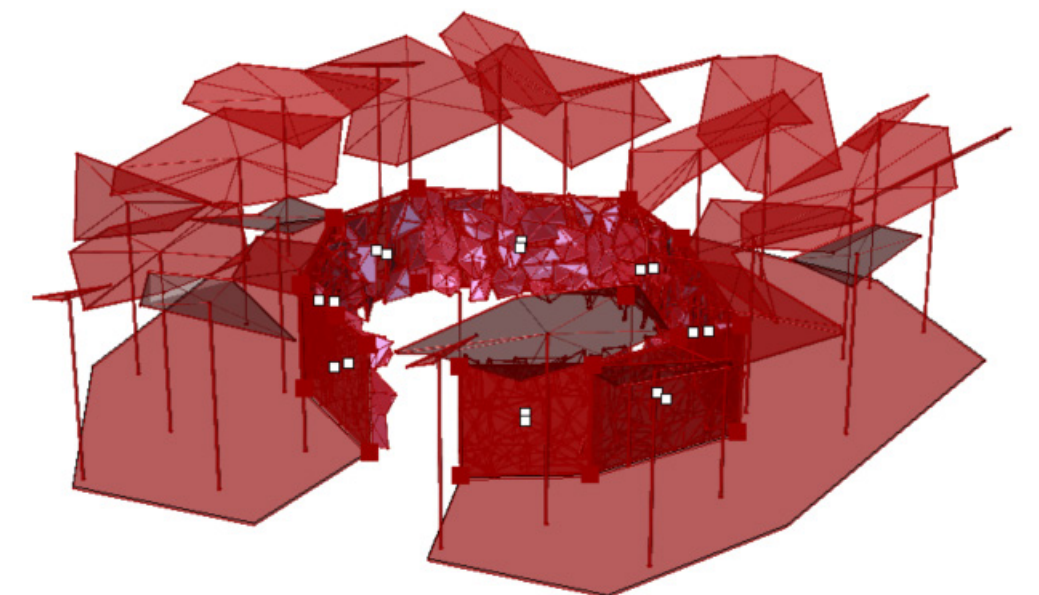
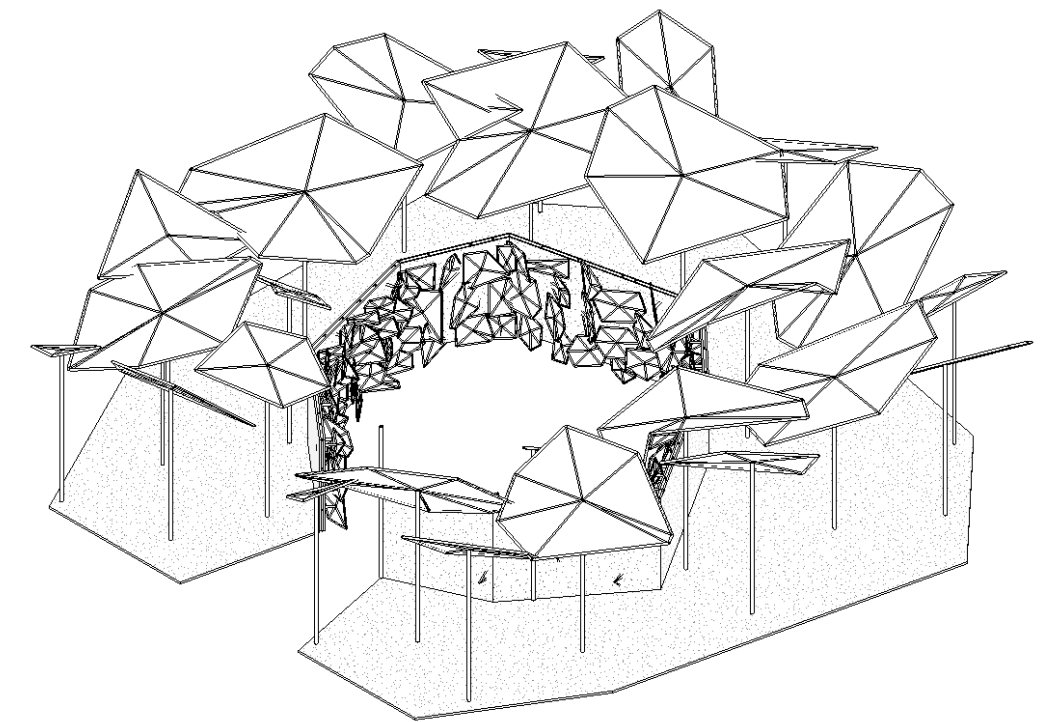
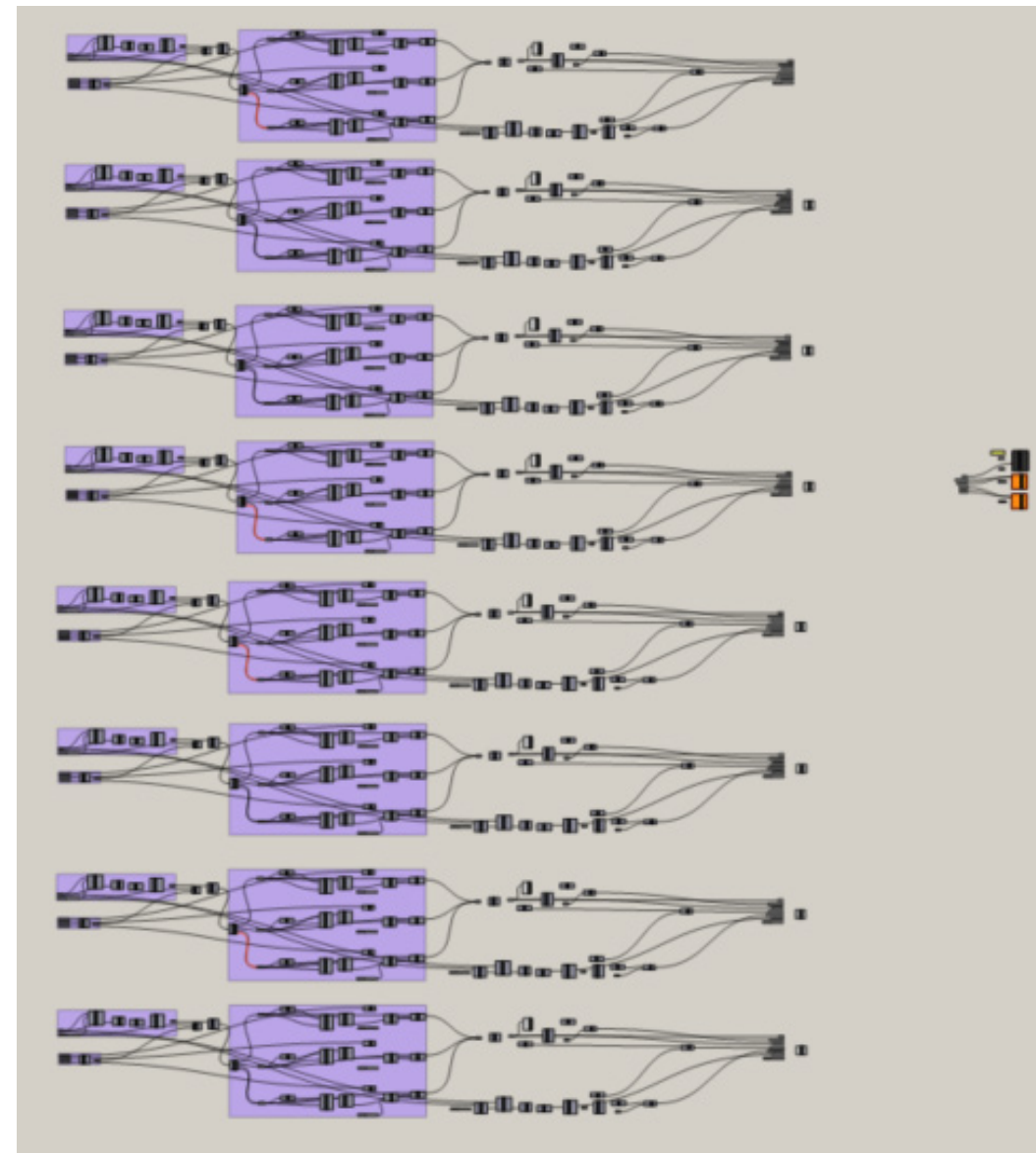
Pipe

6

Rhino Inside

The Pavilion

For the Rhino Inside Revit portion of this assignment, I used a combination of native Revit elements created using boundaries, curves, combined with structural beams that make up the structure and a geometry direct shape that allows for the panels to be visualized.



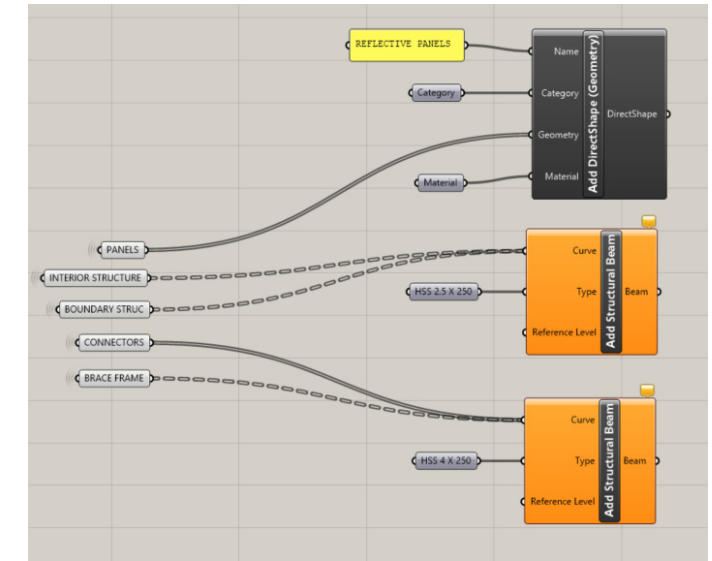
Rhino Inside

The Script



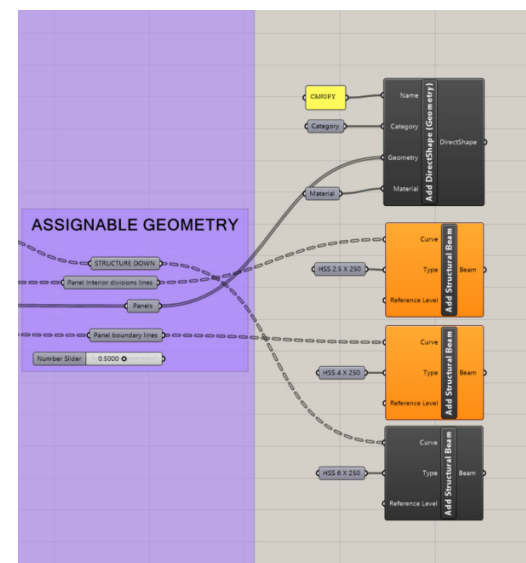
Reflection Space-Overall

To create the reflection space surrounding the center of the pavilion, I used walls as the base surface and then assigned individual rectangular reflection files to each surface. This allowed me to direct the different points towards the centers of opposing surfaces to create multiple kinds of reflections.



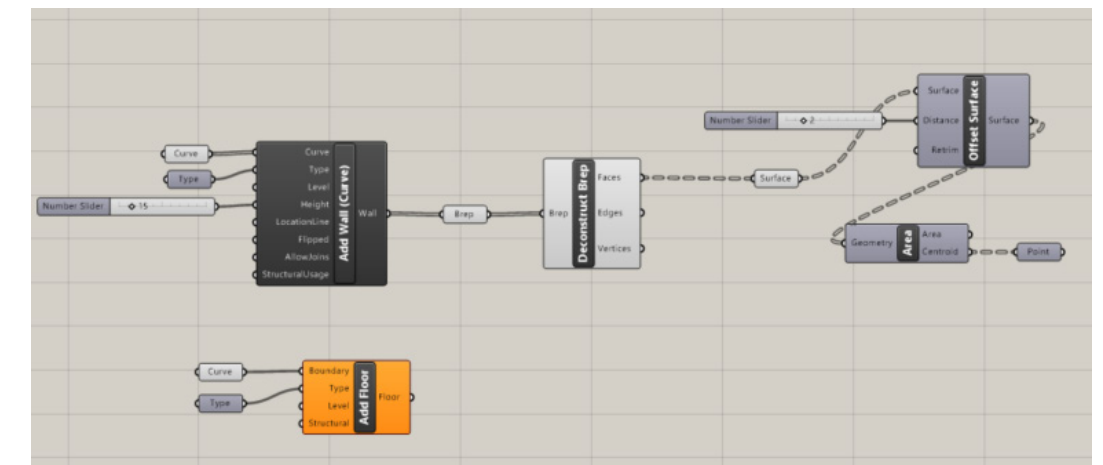
Reflection Space - Revit Process

The main elements from each surface script were all combined into the different categories for ease of attachment. For the structural elements of the reflection wall a structural framing family was used to translate the lines into structural members. The panels were imported via geometry direct shape using a unique name.



Canopy Revit Process

The canopy system similar to the structural wall utilized structural framing families for the structural elements of the canopies, the different elements were separated to reflect more accurate structural constraints.



Walls & Floor

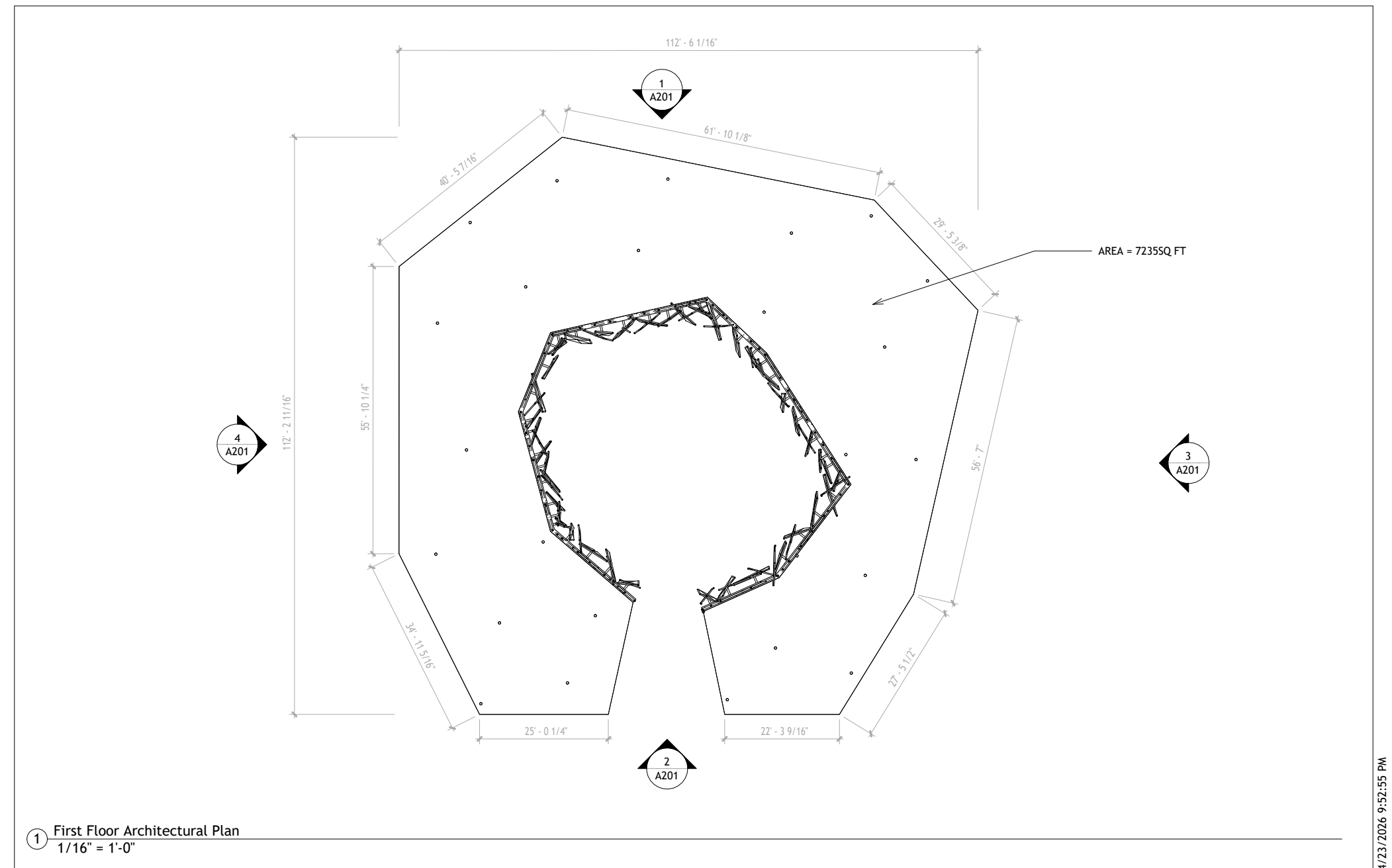
The base Revit geometry was created using boundary curves created in rhino and were translated into native Revit elements. This allowed for the walls and floors to be more easily moved as the design progressed.

DRAWINGS

FLOOR PLAN

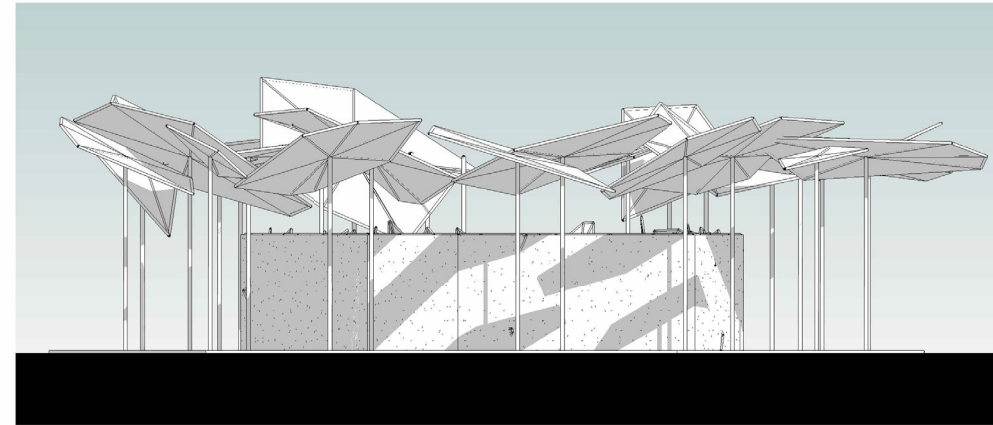
Producing drawings is significantly easier using the Revit workflow, with the use of rhino inside Revit. These plans, elevations, sections and annotated 3D views can be created quickly and customized with pre-made view templates.

In the slides below, examples of these view templates and the ease of documentation is displayed through sections, elevations, and 3D views that can be updated seamlessly as the rhino file develops.

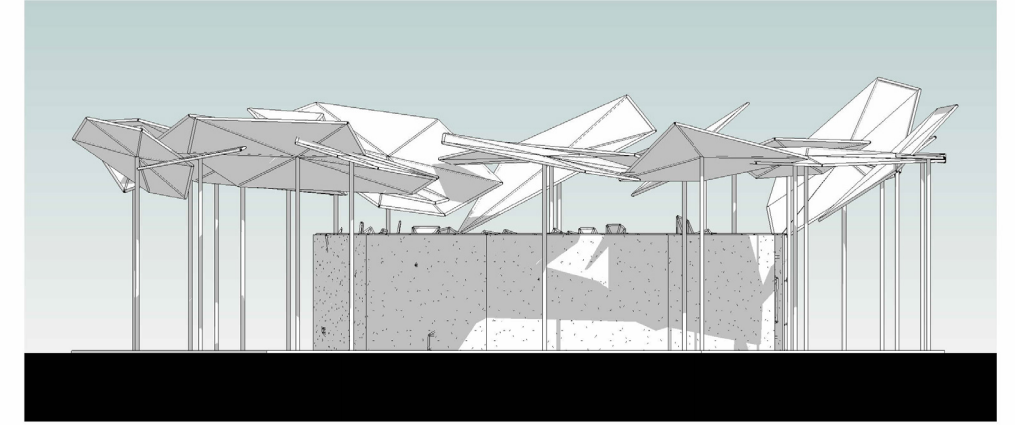


DRAWINGS

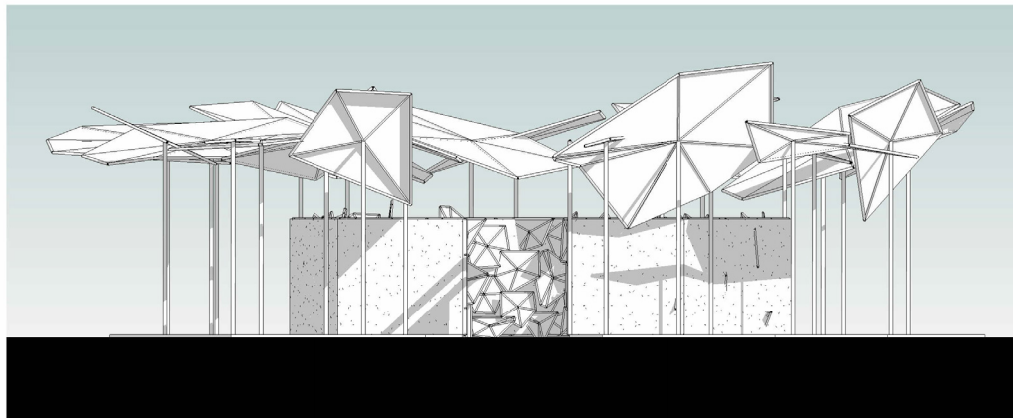
ELEVATIONS



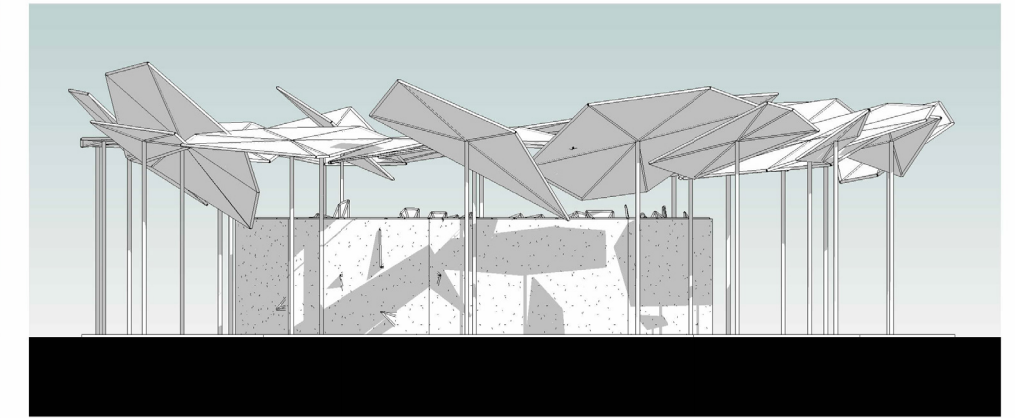
① North Architectural Building Elevation
1/16" = 1'-0"



④ West Architectural Building Elevation
1/16" = 1'-0"



② South Architectural Building Elevation
1/16" = 1'-0"

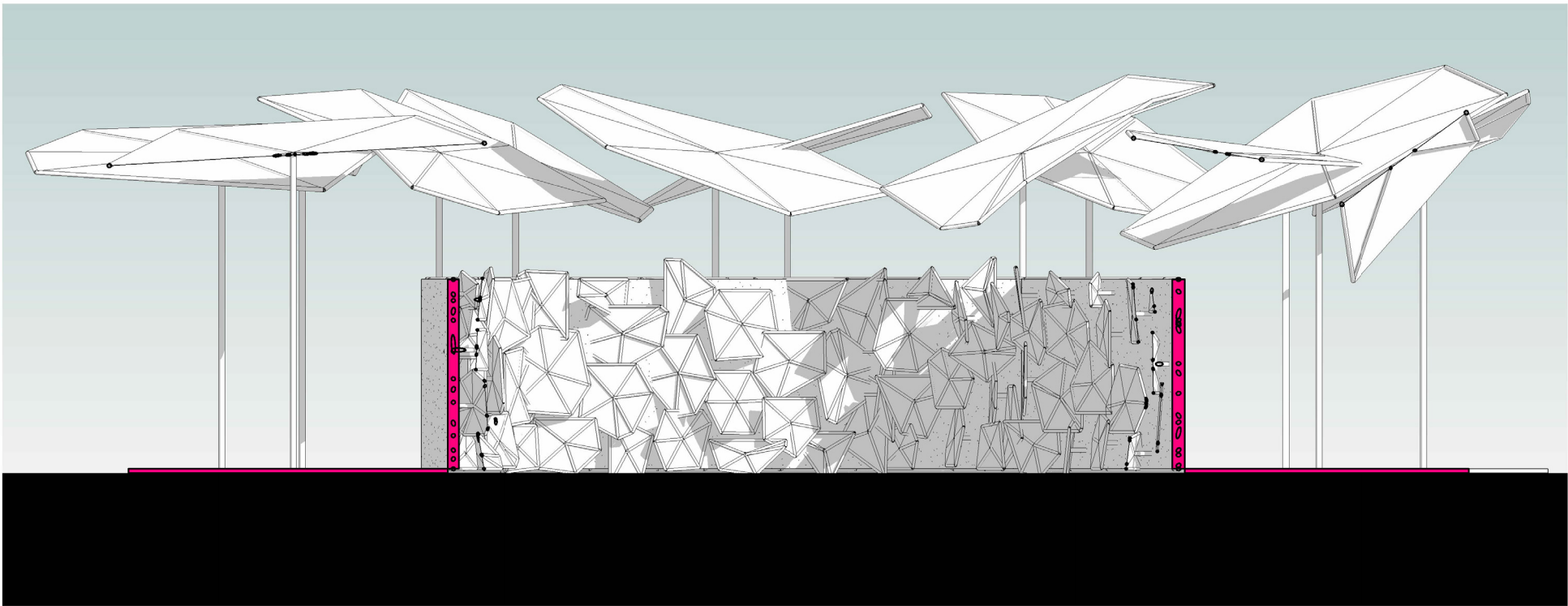


③ East Architectural Building Elevation
1/16" = 1'-0"

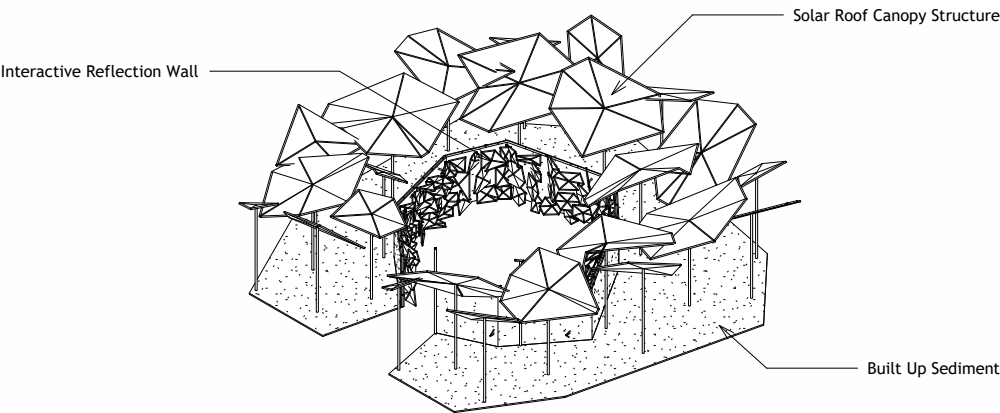
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DRAWINGS

SECTION & 3D



① Section 1
1/8" = 1'-0"



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RENDERING

BIRDS EYE VIEW

These renderings were taken of the more developed studio project that I have worked on in the last few months. The grasshopper through rhino inside process was used throughout this project but for the final visualization, I wanted to take advantage of the skills we were learning in this class to apply towards studio.

Twin motion allows for greater control of the different layers of the renderings. For example, the rendering on the right was to create a sense of the transition from day to night in the desert. After the sun has set the mountains will get dark and only the reflective surface of the salt water flats will remain bright. These settings for exposure, fog layers, and different saturation values allows for more control over the final product.

The next series of renderings were meant to capture the idea of the Oasis in the desert. These spaces would be far to reach but once there, the users would be presented with opportunities to reflect on what it means to be a nomad.



RENDERING

EXTERIOR VIEW



RENDERING

INTERIOR VIEWS

